

Software Installation Request Management using ServiceNow – Capstone Project Documentation

1. Introduction

Project Title: Software Installation Request Management using ServiceNow

2. Project Overview

Purpose

The Software Installation Request Management system is developed to automate the software request and installation process within an organization using the ServiceNow platform. The application minimizes manual effort by allowing employees to submit software requests through a Service Catalog, enabling managers to approve or reject requests, automatically assigning installation tasks to IT administrators, and notifying users throughout the request lifecycle.

Features

- Software Request through Service Catalog
- Manager Approval Workflow
- Automatic Task Creation for IT Team
- Software Installation Tracking
- Email Notifications
- Request Status Monitoring
- Automated Fulfillment Process
- Dashboard and Reporting Support

Project Description

The Software Installation Request Management System is a workflow automation application built on the ServiceNow platform to simplify and standardize software installation requests across an organization.

Employees can request software such as Microsoft Office 365, Visual Studio Code, Google Chrome, Adobe Acrobat Reader, and other approved applications through a user-friendly Service Catalog form. The request captures details including software name, business justification, and requester information.

Once the request is submitted, the system automatically creates a software request record and initiates an approval workflow. The employee's manager receives an approval request and can approve or reject it directly from the ServiceNow platform.

If the request is approved, the application automatically generates an IT fulfillment task and assigns it to the System Administrator responsible for software installation. The administrator installs the requested software, updates the fulfillment status, records completion details, and closes the task.

Throughout the process, automated email notifications keep the employee informed whenever the request is received, approved, rejected, or completed.

The project demonstrates how ServiceNow's Service Catalog, Flow Designer, Business Rules, Notifications, and Task Management modules can be integrated to automate an enterprise software request process efficiently.

3. Architecture

Frontend

- ServiceNow Service Portal
- Native ServiceNow Forms
- Service Catalog

Backend

- Flow Designer
- Business Rules
- Client Scripts
- Notifications
- Task Management

Database

Custom Table:

Software Request Table (u_software_request)

Stores:

- Software Name
- Business Justification
- Request Date

- Requested By
- Manager
- Approval Status
- Fulfillment Status
- IT Notes
- Fulfilled Date
- Related RITM

Architecture Description

The Software Installation Request Management System follows a layered architecture that separates user interaction, workflow automation, business logic, and database management.

1. Presentation Layer

The Presentation Layer consists of the Service Catalog and Service Portal where employees submit software installation requests. Users select the required software, provide a business justification, and submit the request through a structured form.

Client Scripts provide field validation, mandatory field checks, and improve the overall user experience.

2. Application Logic Layer

The Application Layer contains the core automation logic implemented using Flow Designer and Business Rules.

This layer performs:

- Request creation
- Manager approval
- Task generation
- Fulfillment processing
- Status updates
- Notification triggering

Flow Designer automates the request lifecycle without requiring manual intervention.

3. Data Layer

All request information is stored in the custom Software Request table.

The database stores:

- Software Name
- Employee Details
- Manager Details
- Approval Status
- Fulfillment Status
- Installation Notes
- Request Date
- Completion Date

This structured storage enables reporting, auditing, and future reference.

4. Notification Layer

The Notification Engine automatically sends email notifications whenever:

- Request Submitted
- Request Approved
- Request Rejected
- Software Installation Completed

This ensures that employees remain informed throughout the process.

5. Task Management Layer

After manager approval, the application automatically creates an IT Task and assigns it to the System Administrator.

The administrator performs software installation, updates work notes, changes task state to Closed Complete, and updates the fulfillment status.

This automation eliminates manual task assignment and improves operational efficiency.

Overall Workflow

Employee



Submit Software Request



Software Request Record Created



Manager Approval



Approved



IT Task Created



Software Installation



Fulfillment Completed



Email Notification Sent



Request Closed

4. Setup Instructions

Prerequisites

- ServiceNow Personal Developer Instance (PDI)
- System Administrator Role
- Flow Designer Plugin Enabled
- Service Catalog Plugin Enabled
- Email Notification Plugin Enable

Installation Steps

Step 1: Create Custom Application

- Navigate to **System Applications → Studio**
- Create a new scoped application named **Software Installation Request Management**
- Configure the application scope and application menu.

Step 2: Create Software Request Table

Navigate to:

System Definition → Tables

Create a custom table named:

Software Request (u_software_request)

Create the following fields:

- Software Name
- Business Justification
- Requested By
- Manager
- Approval Status
- Fulfillment Status
- Assigned To
- Installation Notes
- Request Date
- Fulfilled Date

Step 3: Create Service Catalog Item

Navigate to:

Service Catalog → Maintain Items

Create a Catalog Item named:

Software Installation Request

Add the following variables:

- Software Name
- Business Justification
- Requested For
- Department
- Priority

Ensure all mandatory fields are configured properly.

Step 4: Configure Client Scripts

Navigate to:

Catalog Client Scripts

Create Client Scripts for:

- Mandatory field validation
- Business justification validation
- Dynamic field behavior
- Prevent empty submission

Step 5: Configure Flow Designer

Navigate to:

Flow Designer

Create a flow that performs the following:

- Trigger when Software Request is created
- Send Approval to Manager
- Wait for Approval Response
- If Approved:
 - Create IT Fulfillment Task
 - Assign Task to System Administrator
 - Update Fulfillment Status
- If Rejected:
 - Update Request Status
 - Send Rejection Email

Step 6: Configure Notifications

Navigate to:

System Notification → Email → Notifications

Create notifications for:

- Request Submitted
- Request Approved
- Request Rejected

- Installation Completed

Recipients should automatically include the requester.

Step 7: Configure Reports

Navigate to:

Reports → Create New Report

Generate reports for:

- Total Requests
- Approved Requests
- Rejected Requests
- Pending Requests
- Completed Installations

Create a dashboard for monitoring request statistics.

Step 8: Testing the Application

Submit a sample request and verify:

- Request Record Creation
- Manager Approval
- IT Task Creation
- Fulfillment Status Update
- Email Notification Delivery
- Task Closure

Once all components are configured successfully, the Software Installation Request Management application is ready for deployment.

5. Folder / Component Structure

The project follows a layered ServiceNow architecture.

1. Service Catalog Layer

Contains the Software Installation Request catalog item.

Responsible for:

- Employee Request Form

- Software Selection
- Business Justification
- Request Submission

2. Client-Side Layer

Implemented using:

- Catalog Client Scripts
- UI Policies

Responsibilities:

- Mandatory Validation
- Dynamic Form Behaviour
- User Input Validation
- Error Messages

3. Workflow Layer

Implemented using:

- Flow Designer

Responsibilities:

- Approval Process
- Task Creation
- Status Updates
- Email Notifications

4. Business Logic Layer

Implemented using:

- Business Rules
- Script Includes

Responsibilities:

- Backend Automation
- Data Validation
- Fulfillment Processing

- Record Updates

5. Data Layer

Uses:

Software Request Table

Stores:

- Request Information
- Approval Status
- Fulfillment Details
- Installation Notes
- Completion Information

6. Notification Layer

Responsible for:

- Request Submitted Email
- Approval Email
- Rejection Email
- Completion Email

7. Reporting Layer

Built using:

- ServiceNow Reports
- Dashboards

Provides:

- Request Statistics
- Pending Requests
- Approval Reports
- Fulfillment Reports

Overall Architecture

UI Layer



Service Catalog



Flow Designer



Business Rules



Software Request Table



Notifications



Reports & Dashboard

6. Running the Application

Step 1

Open the ServiceNow Service Portal.

Navigate to:

Software Installation Request

Step 2

Fill the request form.

Enter:

- Software Name
- Business Justification
- Requested For
- Department
- Priority

Client Scripts validate all inputs before submission.

Step 3

Submit the request.

The application automatically:

- Creates Software Request Record
- Stores Request Information
- Starts Approval Workflow

Step 4

Manager Approval

The manager receives an approval request.

The manager can:

- Approve
- Reject

Step 5

IT Fulfillment

If approved,

The system automatically:

- Creates Fulfillment Task
- Assigns Task to System Administrator

The administrator installs the software and updates the request.

Step 6

Completion

After installation,

The administrator updates:

- Fulfillment Status = Completed
- Fulfilled Date
- Installation Notes

Step 7

Notification Delivery

The requester automatically receives an email confirming that the software installation has been completed successfully.

Step 8

Monitoring

Administrators can monitor the request through:

- Software Request Table
- Task Records
- Flow Execution
- Reports
- Dashboards

This completes the end-to-end software installation request lifecycle using ServiceNow.